

Paper A — Extraoral Temporalis Optical Myography (Draft)

Extraoral Temporalis Optical Myography at 50 Hz: Research Kit, Pipeline, and Feasibility — Draft (Paper A)

PDF: PAPER_A_IEEE_TEMPORALIS_OMG_DRAFT.pdf · HTML

Status: Working draft for co-author review (Koorosh Nabavi / McGill-RI-MUHC · Dr Edward Owens · Dr Pedro Mayoral Sanz · John A. Cogan)

As at: 12 Aug 2026 · Pack 1.1.64 · Research Kit + Dual A SpO₂nEMG nest + measured Mac Dual A 20260812_085110 + research EDF+

Intended venue: [PLACEHOLDER — OWNER: Koorosh] IEEE JBHI / EMBC / Sensors / other — shortlist 2–3

Format note: Markdown for markup; convert to IEEEtran after venue lock.

Review cover: PAPER_A_REVIEW_COVER.md

Feasibility protocol: PAPER_A_FEASIBILITY_PROTOCOL.md · kit

ORALABLE_RESEARCH_KIT.md · placement

TEMPORALIS_ANATOMY_AND_PLACEMENT.md

Alignment audit: PAPER_A_VALIDATION_AND_FUTURE_WORK.md

Canonical collab truth: COLLAB_NABAVI_MCGILL.md · lit distill

LITERATURE_AND_PRIOR_ART.md · figures ../FIGURES.md · Ed/Pedro SB theory ED_PEDRO_SB_FEP_DRAFT_PAPER.md

Claim tone (locked for this draft): Stage A **measurement / methods + small feasibility** paper. Device-inferred overnight states, Dual A concordance, and burden metrics are **engineering phenotypes**, not clinical diagnoses of sleep bruxism, OSA, or disease. No AHI claims. No claim of superiority to Bruxoff. No FDA/CE clearance claims. Patent: “US provisional filed / patent pending” only — **no claim text**.

Title (working)

Oralable Research Kit for Extraoral Temporalis Optical Myography: A 50 Hz Pipeline, Dual sEMG Concordance Path, and Beacon Feasibility (n≈5)

[PLACEHOLDER — OWNER: All] Final title after venue lock.

Authors and affiliations (order TBD)

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[TBD]	<i>Additional co-authors</i>	[PLACEHOLDER]	[PLACEHOLDER]

Corresponding author: [PLACEHOLDER – OWNER: John + Koorosh]

[PLACEHOLDER — OWNER: Koorosh] Exact McGill / RI-MUHC departmental line for byline.

[PLACEHOLDER — OWNER: All] Authorship order and CRediT contributions after first call.

Conflicts / hats: Phase 1 academic work runs through **McGill / RI-MUHC**. A later Dianyx path (if any) does not change Paper A inventorship. Nabavi's Dianyx CIO role goes in COI (see § Conflicts).

Abstract

Sleep bruxism (SB) assessment still leans on polysomnography with audio-visual recording (PSG-AV). Ambulatory tools are mostly surface EMG (GrindCare, Bruxoff), mandibular motion, acoustic home sleep tests (AcuPebble), or intraoral force sensing. Recent reviews list few **extraoral temporalis** optical ambulatory devices. We describe the **Oralable Research Kit**: extraoral temple PPG plus accelerometry (MAM) seated on the **anterior temporalis** elevating belly, optional temporalis sEMG (ANR M40) for Dual Protocol A concordance, and a **50 Hz**

optical myography (OMG) pipeline. MAM senses hemodynamic occlusion (IR-DC) and vitals; ANR senses electrical motor activity of the same belly. IR-DC occlusion features feed a **state hypnogram** and engineering burden metrics over multi-hour wear (1–6 h+). A Beacon feasibility protocol ($n \approx 5$) nests Pedro Mayoral's 1–2 h oxygen / MAD arm with AcuPebble AHI context. We do not claim AHI equivalence.

[PLACEHOLDER — OWNER: John] Results sentences — fill after pilot N , e.g.:

"In $N=[]$ Research Kit participants, $[]\%$ of sessions exported CSV; Arm P windows $n=[]$; Dual A Mac sessions $n=[]$; ≥ 6 h nights $n=[]$."

Index terms—photoplethysmography, optical myography, temporalis, sleep bruxism, wearable sensors, surface EMG, overnight monitoring, SpO₂, accelerometry.

I. Introduction

Sleep bruxism is repetitive masticatory muscle activity during sleep. Clinicians often judge masseter and temporalis patterns against PSG-AV. Ambulatory devices are mainly screening tools until validation improves [1].

Through late 2024, commercial ambulatory systems are mostly **sEMG**, Type II PSG, mandibular movement, or **intra-splint force** [1]. That leaves room for a small **extraoral** optical sensor on the anterior temporalis—outside the mouth—for multi-night home wear.

Prior jaw-adjacent optical work includes (i) **intraoral** PPG for sleep cardiorespiratory monitoring [2], and (ii) **awake** chewing detection with ear-hook PPG, audio, and accelerometry [3]. Neither targets overnight **temporalis OMG** via IR-DC hemodynamic occlusion for jaw-load phenotyping.

This Paper A contributes:

1. A system description of the **Oralable Research Kit** (Gen1 temple MAM + ANR M40 sEMG + iOS/Mac capture paths, 50 Hz research grid), including anterior temporalis seat rationale.
2. A signal-processing pipeline for heart-rate band PPG, IR-DC occlusion features, overnight state classification, and Dual Protocol A alignment vs temporalis sEMG (optical vs electrical paths on the same belly).

3. An overnight **state hypnogram** as the primary at-a-glance map for multi-hour wear (1–2 h Arm P; stretch ≥ 6 h).
4. A Beacon feasibility protocol ($n \approx 5$) with Pedro Mayoral's oxygen / MAD arm and a descriptive competitor landscape (Bruxoff, GrindCare, AcuPebble).

Paper **B** (clinical phenotype / intake labels) and deeper PSG-AV concordance are future work (§IX). Dual A / ANR descriptive concordance is **in scope for methods + precursor results** in this rewrite; diagnostic SB claims are not.

Scope note (Research Kit): Field sessions start with temple HR/SpO₂ and wear QC on Gen1 hardware. The same kit supports Dual Protocol A (Mac) and longer iOS wear. IR-DC jaw-load and Dual A outputs are engineering phenotypes. Research Kits stay **gated / not yet shipped** until charge-to-temple clears; target **5 kits to Pedro by 31 Aug 2026** [8].

II. Related Work

A. Ambulatory sleep bruxism and related devices

Li *et al.* review ambulatory SB devices. EMG-only tools can overestimate SB without audio-visual context. Several commercial tools report partial PSG-AV comparisons with mixed sensitivity and specificity [1]. Their commercial table lists **no temple optical / PPG-OMG ambulatory device**—the gap this Research Kit targets.

Device / class	Modality	Primary outputs	Role vs Oralable
ANR M40	Temporalis research sEMG	EMG amplitude / bouts	In-kit Dual A comparator
AcuPebble	Neck acoustic HSAT $\pm$ SpO ₂	AHI / ODI	Pedro's OSA reference — nest, do not replace [13]
GrindCare	Anterior temporalis sEMG	Muscle activity / biofeedback	Peer class at same site; Oralable is optical OMG
Bruxoff	Masseter sEMG + ECG	Ambulatory SB-oriented counts	Reference ambulatory EMG (not PSG gold) [12]
Happy Ring	Finger PPG ring (Oura-like)	hAHI (FDA MNR SaMD)	Same HSAT-output class; not temporalis / SB [16]
Oralable MAM	Temple PPG + ACC	HR, SpO ₂ , IR-DC OMG, hypnogram	Extraoral optical jaw-load + vitals

Cid-Verdejo *et al.* compare Bruxoff to PSG in OSA cohorts and report overestimation risk [12]. Bruxoff is a useful ambulatory reference, not a diagnostic gold standard.

B. Screening questionnaires

Lobbezoo *et al.* describe BruxScreen (patient questionnaire + clinical form) as a lighter alternative to full STAB assessment for everyday practice [4]. We use BruxScreen as an **optional clinical intake / label** for later phenotype papers, not a substitute for instrumented overnight mapping.

C. Adjacent optical mastication sensing

Papapanagiotou *et al.* detected chewing with ear PPG, audio, and ACC in awake settings [3]. That shows PPG near the jaw can carry masticatory information. Our goal differs: overnight SB-related load vs nutrition / chewing; temple temporalis vs ear; IR-DC occlusion / TFI vs SVM late fusion.

D. Prior intraoral PPG (authors' related history)

Nabavi, Cogan, Roy, *et al.* reported sleep monitoring with **intraorally** measured PPG [2]. Paper A is anatomically distinct: **extraoral temporalis** clip sensing. Intraoral appliance sensing is out of scope for this collaboration's product and paper framing [5].

[PLACEHOLDER — OWNER: Koorosh] Confirm preferred citation form / DOI for [2] and any additional McGill wearables papers to cite.

E. Clinical theory (Beacon co-authors; not methods)

Owens and Mayoral outline an FEP-informed account of SB endotypes and a MAD-selective RMMA hypothesis (airway-linked episodes should fall when MAD/CPAP works; stress- or dopamine-linked episodes should not) [17]. That paper is **Hypothesis and Theory**. It supplies clinical motivation for Pedro's oxygen/MAD arm and for later phenotype work. It is **not** a device, pipeline, or feasibility methods source for this manuscript. Paper A does not test FEP constructs or report homeostatic latency.

III. System and Hardware — Orable Research Kit

A. Kit definition

The **Orable Research Kit** combines (i) Gen1 Orable temple MAM, (ii) ANR M40 temporalis sEMG, (iii) magnetic charge case, and (iv) iOS Orable app for 1–6 h+ BLE collection, with Mac Dual Protocol A for labeled concordance [8], [14], [15].

Kit item	Role
Orable Gen1 clip + case	Temple PPG/ ACC · FW 1.0.70
ANR M40	Temporalis Analog EMG (~10 Hz notify)
iOS Orable 4.3.3	Long-wear vitals / overnight export
Mac Dual A scripts	Shared Protocol A cues · paired logs



Fig. 1. Extraoral temporalis sensing (this work) versus intraoral appliance sensing (out of scope). Asset: FIG-CO-011 — placeholder SVG; redraw for publication.

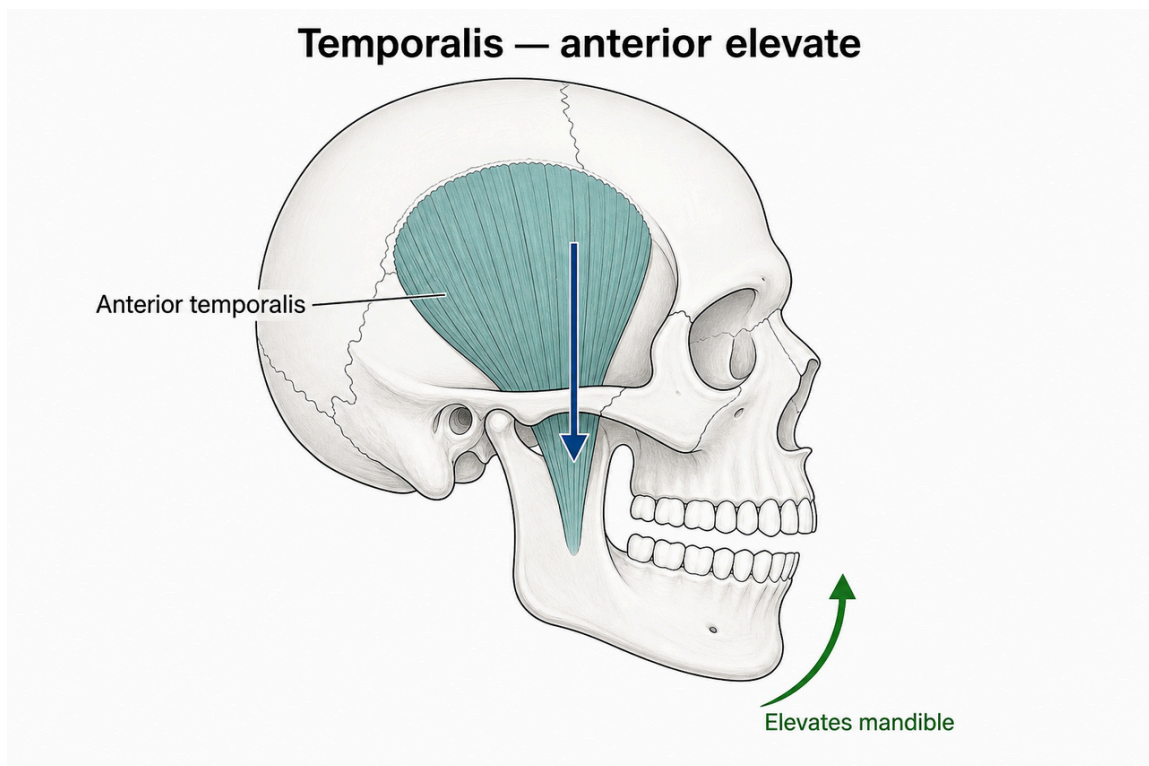
A2. Anatomical target — anterior temporalis

The temporalis is a fan-shaped masticatory muscle arising from the temporal fossa and inserting on the coronoid process. Functionally it has an **anterior** belly with near-**vertical** fibers that **elevate** the mandible (tooth occlusion / bite) and a **posterior** belly with near-**horizontal** fibers that assist elevation and uniquely **retract** the mandible [15]. The Research Kit seats both Orable and ANR on the

anterior elevating belly (clench bulge between the eyebrow tail and the ear)—the same site class as anterior-temporalis sEMG peers (e.g. GrindCare)—not on the posterior retracting belly [14], [15].

Device	Physics on the anterior belly	Primary derived quantities
Oralable MAM	Optical PPG (+ ACC): muscle stiffening compresses local perfusion	IR-DC / OMG troughs; HR ; SpO₂ ; SASHB; motion / sync taps
ANR M40	Surface EMG (Analog 0x2A58)	Raw amplitude; bout onset / offset for Dual A concordance

EMG bout onset typically **leads** the IR-DC trough by about 1–5 s (hemodynamic lag). Dual A electrode long axis is **vertical**, parallel to anterior fibers [14].



*Fig. 2(a). Temporalis function — anterior (vertical) fibers elevate the mandible. Oralable schematic FIG-CO-056 after standard anatomy [15]. **Primary Research Kit** seat for Oralable PPG and ANR.*

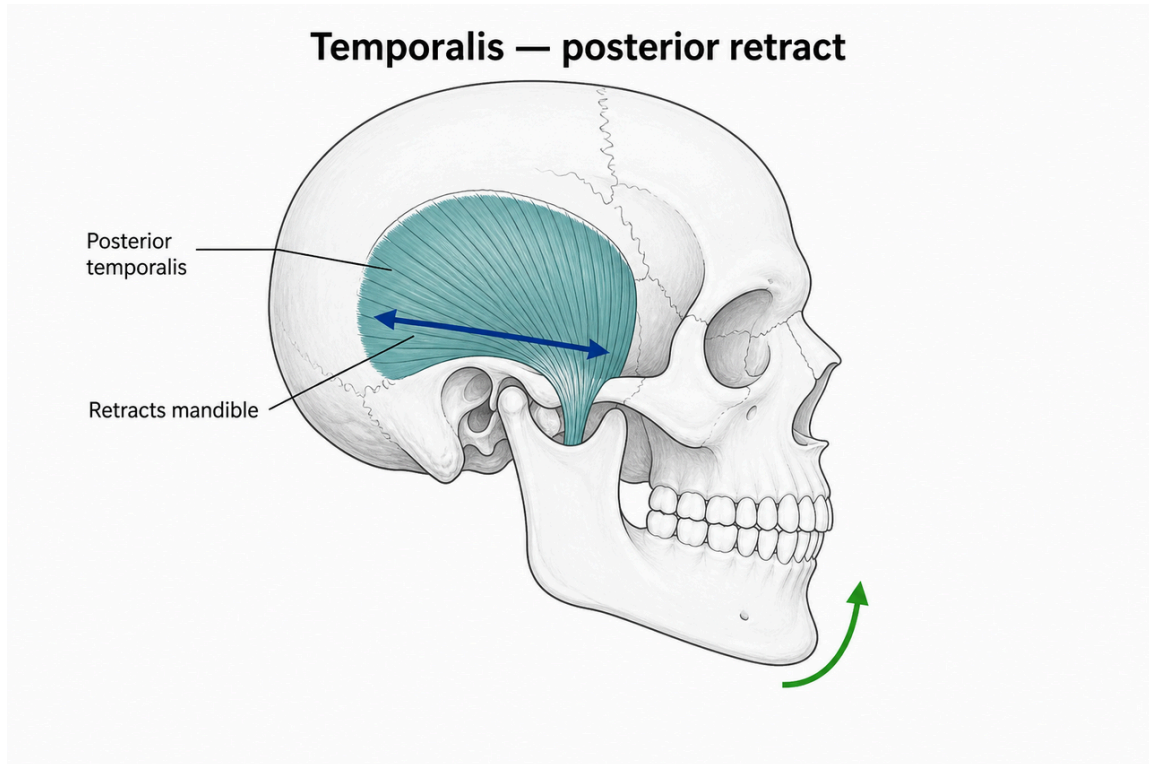


Fig. 2(b). Temporalis function — posterior (near-horizontal) fibers retract the mandible [15] (asset FIG-CO-057). Shown for anatomical contrast; **not** the primary Dual A seat.



Fig. 2(c). Temple placement schematic (practice overlay). Asset: FIG-CO-003 — placeholder; align marks with Fig. 2(a) anterior belly.



*Fig. 2(d). Gen1 Orable device. Asset: FIG-CO-012 — **[PLACEHOLDER — OWNER: John]**.*



*Fig. 2(e). Research Kit contents. Asset: FIG-CO-016 — **[PLACEHOLDER — OWNER: John]**; photo path also FIG-CO-016 PNG / FIG-CO-049.*



Fig. 2(f). ANR M40. Asset: FIG-CO-026 — prefer photo FIG-CO-026 PNG when publishing.



*Fig. 2(g). Dual A practice: ANR long-axis **vertical** on anterior temporalis over / pressing the Oralable optical stack (asset FIG-CO-031). Related: FIG-CO-054 hybrid · FIG-CO-055 finger-press placement.*



Fig. 2(h)–related. Bruxoff / AcuPebble / GrindCare photo slots: FIG-CO-027 / 028 / 029 — [PLACEHOLDER — OWNER: Pedro / John].

B. Sensing and electronics (summary)

Item	Draft specification
PPG	Maxim MAXM86161 (multi-wavelength; Red / Green / IR used in pipeline)
Accelerometer	LIS2DTW12 (actigraphy / sync taps / motion context)
MCU / radio	Kaga ES2832AA2 (nRF52832) Gen1
BOM / PCB	Gen1 · BOM REV8 · PCB REV10
Stream	BLE notify path; research grid 50 Hz (20 ms) after resampling
Firmware (pilot)	1.0.70 ship (min gate 1.0.63)
Companion app	iOS Orable patient app 4.3.3 (vitals + overnight report path)
Dual A sEMG	ANR M40 Automation IO Analog EMG 0x2A58

[PLACEHOLDER — OWNER: John] Add a concise electrical block diagram from Altium / system architecture — no confidential claim charts.

C. Data products

Raw and derived channels export as time-aligned CSV (50 Hz grid) for offline analysis; the patient app can render an overnight state hypnogram and export a multi-page clinical Temporalis PDF. Dual A produces paired Oralable + ANR logs and concordance packs via `align_anr_oralable_concordance.py`. Methods parity targets exist between Python research tooling and Swift production (OralableCore) [6].

IV. Signal Processing Methods

```
flowchart LR
    Raw[Raw[BLE_or_CSV_raw]] --> Resample1[Resample[Resample_50Hz]]
    Resample1 --> BP[BP[Butterworth_0p5_to_8Hz_green]]
    Resample1 --> LP[LP[IR_DC_lowpass_under_1Hz]]
    Resample1 --> Acc[Acc[Accel_actigraphy]]
    BP --> HR[HR[Heart_rate]]
    LP --> Occ[Occ[Occlusion_trough]]
    Acc --> Sync[Sync[Sync_taps]]
    Occ --> Metrics1[Metrics[TFI_SASHB_states]]
    HR --> Metrics2[Metrics]
```

Fig. 3(a). Processing overview (methods draft).

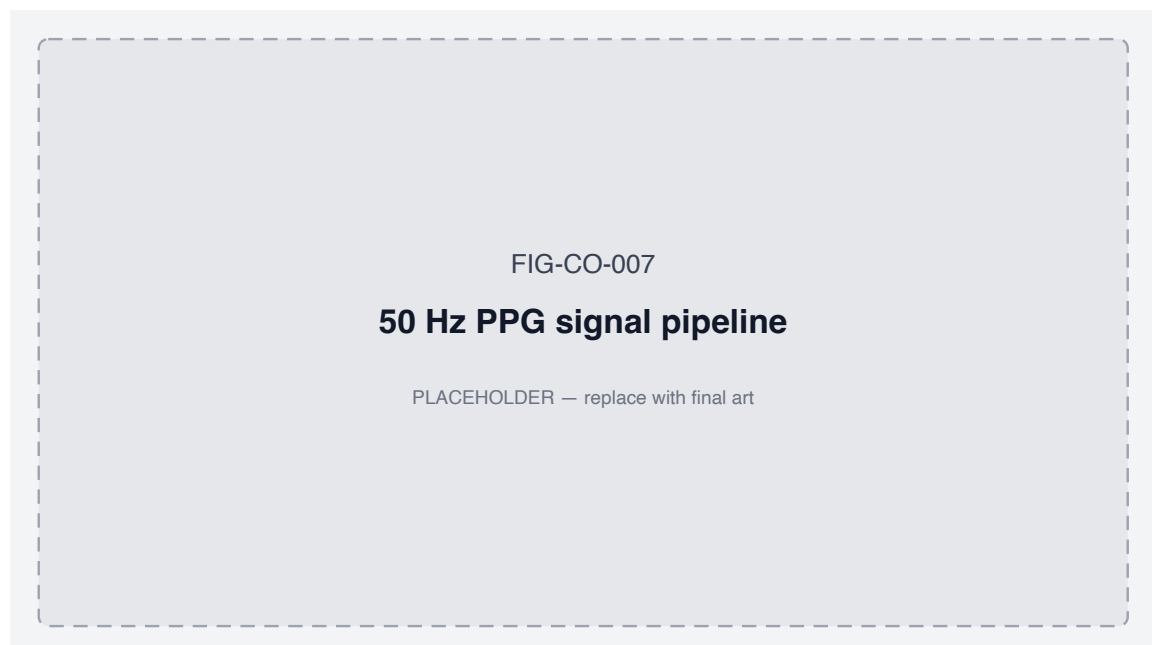


Fig. 3(b). 50 Hz PPG pipeline figure slot. Asset: FIG-CO-007 — replace with publication redraw.

A. Resampling

All analysis channels are linearly interpolated onto a strict **50 Hz** grid (20 ms). Accelerometer samples acquired at higher rates are synchronized to the same grid for overnight fusion.

B. Heart-rate PPG path

Green (or configured AC) channel: Butterworth bandpass approximately **0.5–8.0 Hz** for pulsatile analysis and heart-rate estimation under temple coupling. SpO₂ estimation uses red/IR ratio methods with quality gating.

[PLACEHOLDER — OWNER: Koorosh + John] Exact filter order, SOS coefficients, and SpO₂ calibration coefficients for the paper table (from shared algorithm spec).

C. IR-DC occlusion / OMG path

Infrared DC is low-pass filtered (<1 Hz) to extract slow hemodynamic shifts associated with temporalis load (optical myography / occlusion troughs). Clench-related detections are cross-checked against IR-DC trough depth [6].



Fig. 4. IR-DC occlusion trough illustration. Asset: FIG-CO-006 — [PLACEHOLDER — OWNER: John] replace with real annotated bout from Protocol A / overnight log.

C2. Dual Protocol A concordance (ANR M40)

Labeled Dual Protocol A runs Orable and ANR M40 in parallel on the **anterior temporalis** under shared wall-clock cues (5-tap sync at Protocol A minute 01:00) via Mac scripts [14], [15]. ANR Red Dots are oriented with long axis **vertical** (parallel to anterior fibers); an EMG amplitude preflight (rest → hard clench → rest) gates contact quality before cues. Post-hoc alignment resamples EMG and optical features onto the **50 Hz** grid for descriptive bout timing and amplitude comparison (electrical lead vs IR-DC lag). The same pack nests Orable SpO₂ / SASHB with ANR EMG bouts (desat events lasting ≥10 s; co-occurrence within ±5 s) as an **AcuPebble-style oxygen-burden context** — not claimed ODI/AHI and not Bruxoff equivalence [13], [14]. Align also writes a research **EDF+** (`session.edf`) with ANR EMG when Dual A is used; iOS Dual A Share can include the same. Mac remains the methods reference until iOS parity is proven. This is a **Paper A methods path and precursor result**, not PSG-AV SB diagnosis.

Engineering precursor (12 Aug 2026): Mac Dual A session 20260812_085110 completed (EMG preflight gate 70; clench max 83). Concordance pack `plots/concordance/20260812_085110/` includes overlay, `NEST.md`, and `session.edf`. Temple SpO₂ was computed (aligned mean ≈ 89.5%); engineering SASHB ≈ 929 %·s. Median EMG→IR-DC lag ≈ 4.9 s. Bout F1 vs Protocol A labels was 0 on this pack — placement/QC follow-up, not partner N. A state-hypnogram **layout** pack from the same Dual A gold is at `plots/overnight_report/TEMPORALIS_20260812_085110_dualA/` (wear ≈ 6.0 min — not an evaluable overnight).

D. Session metrics (engineering)

Metric	Definition (draft)	Notes
TFI	Temporalis Fatigue Index, 0–100 session score from IR-DC / green AC slopes	Engineering load index
SASHB	Engineering SpO ₂ <90% area (%·s): $\Sigma (90 - \text{SpO}_2) \cdot dt$	Rate = total ÷ wear hours; not Azarbarzin event-linked HB; not AHI
SpO₂∩EMG nest	Fraction of ANR EMG bouts co-occurring with SpO ₂ < 90% ≥10 s (±5 s)	Dual A pack (<code>NEST.md</code>); not ODI/AHI
Research EDF+	50 Hz Dual A time series (+ ANR EMG when used)	<code>session.edf</code> — research handoff; not PSG
Rescue rate	Rescue-class events ÷ wear hours	Device-inferred; not AHI

Metric	Definition (draft)	Notes
Activity mix	Tonic minutes ÷ wear hours (phasic secondary)	Banded for evaluable nights

Band cutoffs (Low / Moderate / High) are **provisional pilot defaults** for UI/reporting and must be recalibrated from ≥ 6 h cohort distributions [7].

E. Overnight state classifier

Epochs are labeled into device-inferred states: **quiet**, **tonic**, **phasic**, **rescue**, **recovery** (implementation: Python `overnight_states` with Swift OvernightStateClassifier parity target) [6], [7].

[PLACEHOLDER — OWNER: Koorosh + John] Publishable state machine / feature table (window length, thresholds, hysteresis). Keep wellness wording.

V. Overnight Representation: State Hypnogram

The primary overnight graphic is a **state hypnogram**—a barcode-style map of quiet / tonic / phasic / rescue / recovery across the night. Layout resembles a sleep-stage hypnogram, but bands show **device-inferred jaw-load / recovery states**, not PSG sleep stages [7].

Evaluable night rule (Paper A / Arm E/J): primary overnight Results and band recalibration require ≥ 6 h worn (goal 8 h). The iOS app may unlock provisional morning-card bands from ≥ 1 h for pilot UX; do not treat 1 h sessions as Paper A overnight N. Protocol A/B minutes are not sleep sessions.

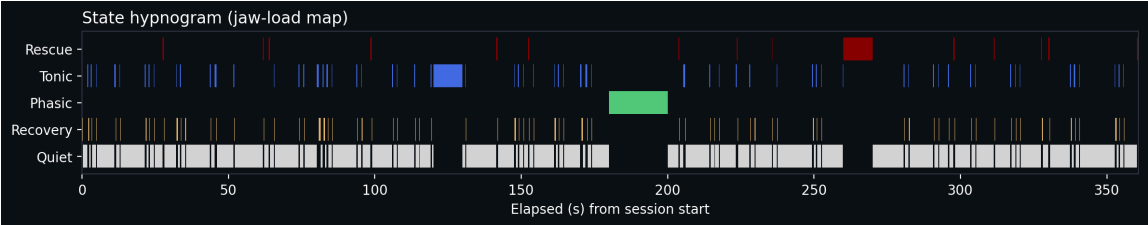


Fig. 5. Illustrative state hypnogram from engineering pack TEMPORALIS_20260724 (asset FIG-CO-025). **Important:** wear ≈ 6.0 min — *not* an evaluable ≥ 6 h overnight. A second layout pack from Dual A 20260812_085110 is at `plots/overnight_report/TEMPORALIS_20260812_085110_dualA/02_state_hypnogram.png` (also ~ 6 min). Replace with consented ≥ 6 h nights before claiming overnight results.



Fig. 5 (supplement). Band-chip layout stub (FIG-CO-019) — optional panel beside hypnogram.

Graphing hierarchy for partner review: (1) state hypnogram, (2) hourly stacked burden + SASHB, (3) IR-DC + SpO₂ dual-rail, (4) event table; 3D cluster appendix only [7].

VI. Pilot Protocol (Beacon · Research Kit · n≈5)

Canonical ops protocol: [PAPER_A_FEASIBILITY_PROTOCOL.md](#) · kit [ORALABLE_RESEARCH_KIT.md](#).

A. Clinical setting

Dental Sleep Medicine partners at **Beacon** (Owens, Mayoral) run Research Kit feasibility: home or clinic wear, chairside review of the overnight map, and Pedro’s MAD titration context nested with AcuPebble AHI when available [8], [13].

B. Structured collection

Protocol / arm	Purpose	Anchor	Duration
Core	Setup, comfort, HR/SpO ₂ QC, export	Temple mount	≥5–10 min
Arm P (Pedro)	Oxygen burden ± MAD	Continuous wear	1–2 h

Protocol / arm	Purpose	Anchor	Duration
Arm E/J	Evaluable overnight hypnogram	Home wear	≥6 h (goal 8 h)
Protocol A / Dual A	Labeled training + ANR concordance	5-tap sync	~6 min Mac Dual A
Protocol B	Validation gates (later)	First 3-tap	~4.5 min

Details: [../TEMPORALIS_COLLECTION_PROTOCOL.md](#).

Optional intake: BruxScreen-Q ($\pm$ BruxScreen-C) [4] for Paper B labels.

Landscape nesting: Bruxoff when available [12]; AcuPebble AHI context [13]; GrindCare as peer class in discussion.

C. Ethics, consent, and data governance

[PLACEHOLDER — OWNER: Ed/Pedro] Beacon / local ethics path, consent language (wellness research vs clinical study), data controller, retention.

[PLACEHOLDER — OWNER: Koorosh] McGill / RI-MUHC REB requirements if analyzing data under McGill affiliation.

[PLACEHOLDER — OWNER: John] JAC data-processing agreement template for partner CSV / PDF sharing (no inventorship transfer).

D. Target sample

Tier	Target	Use
Paper A feasibility	n≈5 Beacon (Pedro $\pm$ Ed first, then +3)	Wear, QC, Arm P, Dual A precursor, landscape
Research Kit ship	5 kits by 31 Aug 2026	Field N gate [14]
Tier 1 (later)	≈ 20–30 users $\times$ 3–5 Protocol A	Core ML cohort [9]
Overnight subset	≥6 h nights	Hypnogram / band recalibration
Deeper EMG / PSG-AV	Later papers	Bruxoff/PSG-AV diagnostic concordance

[PLACEHOLDER — OWNER: All] Lock inclusion/exclusion, demographics stratification (sex, age, habitus, skin tone) [9].

VII. Results

All quantitative results below are placeholders. Do not treat the TEMPORALIS_20260724 exemplar as study results.

A. Cohort (Table I)

Item	Value
Enrolled N	[PLACEHOLDER – OWNER: Ed/Pedro + John]
Evaluable overnight nights (n)	[PLACEHOLDER]
Mean $\pm$ SD wear hours	[PLACEHOLDER]
% nights ≥ 6 h	[PLACEHOLDER]
Demographics	[PLACEHOLDER]

B. Signal quality / coupling (Table II)

Metric	Value
IR-DC coupling in-range fraction	[PLACEHOLDER – OWNER: John]
Green PPG SNR (dB) after BP filter	[PLACEHOLDER – OWNER: Koorosh + John]
SpO ₂ quality-gate pass rate	[PLACEHOLDER]
Dropouts / BLE gaps	[PLACEHOLDER]

C. Overnight phenotype descriptives (Table III) — engineering only

Metric	Median (IQR) or Mean $\pm$ SD
TFI	[PLACEHOLDER]
SASHB / h (%·s / h)	[PLACEHOLDER]
Rescue events / h	[PLACEHOLDER]
Tonic min / h	[PLACEHOLDER]
State-time fractions (quiet/tonic/phasic/...)	[PLACEHOLDER]

D. Illustrative short-session examples (non-inferential)

Two ~6 min engineering packs demonstrate layout / Dual A toolchain. **Neither** is an evaluable ≥ 6 h overnight.

KPI	TEMPORALIS_20260724	Dual A 20260812_085110
Wear	~360.8 s (~6.0 min)	~358.6 s (~6.0 min)
TFI	~47.7	~45.2
SASHB (session total)	~688.5 %·s	~929 %·s
SpO ₂ mean / min	~91.9 / 85.0	~90.5 / 85.0 (overnight clip); aligned min ≈ 6
EMG→IR- DC median lag	—	≈ 4.9 s
Hypnogram layout	FIG-CO-025	.../TEMPORALIS_20260812_085110_dualA/02_s
Research EDF+	—	plots/concordance/20260812_085110/sessi

[PLACEHOLDER — OWNER: John] Replace with a consented ≥6 h night before overnight inferential claims. Do not treat Dual A minutes as partner *N* or as Azarbarzin HB.

E. Engineering readiness (as at 12 Aug 2026 — not partner *N*)

Item	Status
Mac Protocol A + Core ML Tier 0 (TEMPORALIS_20260724)	Done (~6 min eng pack)
Overnight report toolchain + hypnogram layout	Done (layout on eng + Dual A packs)
Mac Dual Protocol A + align + SpO ₂ ∩EMG nest	Measured pack 20260812_085110
Research EDF+ (session.edf with ANR EMG)	Mac default on; iOS Dual A Share
iOS Orable long wear (1–6 h+)	Supported (no hard duration cap)
iOS Dual A overnight / ≥6 h paired	Follow-on
Research Kits with Pedro	Gated — target 5 kits by 31 Aug 2026

F. Agreement endpoints

Comparison	Statistic	Scope
Dual A Orable vs ANR bout timing	Precursor: median lag ≈ 4.9 s (20260812_085110); F1 vs labels = 0 this pack — [PLACEHOLDER – more packs]	Paper A precursor
vs Bruxoff (when available)	Descriptive output classes	Paper A landscape
vs AcuPebble AHI/ODI	Nest only — no AHI claim	Arm P context
vs BruxScreen / dentist impression	[PLACEHOLDER – Paper B]	Later
vs PSG-AV RMMA	[PLACEHOLDER – later]	Later

VIII. Discussion

A. Positioning

This draft treats the **Orable Research Kit** as an **extraoral anterior-temporalis OMG** system for multi-hour jaw-load and vitals mapping [15]. Orable MAM and ANR sense the same elevating belly by different physics (optical occlusion + vitals vs surface EMG). The kit complements PSG-AV diagnosis [1] and AcuPebble AHI [13]; finger HSAT rings such as Happy Ring [16] sit on the same AHI-class shelf. It does not replace them. Bruxoff and GrindCare are the ambulatory EMG peer class [1], [12]. ANR Dual A is the in-kit electrical comparator. The state hypnogram is the proposed primary display for 1–6 h+ home wear.

B. Competitor output comparison (plan)

Device	Typical headline outputs	Orable Kit outputs (this paper)
ANR M40	EMG amplitude / bouts	Dual A alignment vs IR-DC
AcuPebble	AHI, ODI	SpO ₂ burden / wear — not AHI
GrindCare	Temporalis sEMG activity	Optical OMG at same site class
Bruxoff	EMG/ECG ambulatory SB-oriented counts	IR-DC / states / TFI (engineering); optional nest
Happy Ring	hAHI (finger ring HSAT)	SpO ₂ burden / wear — not AHI [16]

C. Limitations

1. No PSG-AV concordance in Paper A (planned later).
2. Dual A concordance is **descriptive**, not diagnostic SB validation.
3. State labels are **device-inferred**; orofacial confounding is possible (as with EMG ambulatory tools [1]).
4. Band cutoffs are provisional.
5. Temple optical coupling depends on **anterior-belly** placement, motion, and skin/hair; QC gates (and Dual A EMG preflight) are required. Posterior temporalis is anatomically real but is not the primary occlusion seat [15].
6. Stage A wellness framing—do not present results as medical diagnosis.
7. Engineering exemplars (~6 min) are not cohort evidence; kits were not shipped as of this draft.

D. Relation to prior intraoral PPG

Extraoral temporalis sensing must stay distinct from prior intraoral PPG [2]. That keeps this McGill academic work separate from intraoral appliance product lines [5].

IX. Future Work — Study Ladder

```
flowchart LR
    PaperA[Paper_A_ResearchKit_feasibility_n5] --> Tier1[Tier1_n_20_to_30]
    Tier1 --> PaperB[Paper_B_phenotype_BruxScreen]
    PaperB --> PaperC[Paper_C_PSG_AV_concordance]
    PaperA --> Overnight[Overnight_hypnogram_bands]
    Overnight --> PaperB
    PaperA --> DualA[Dual_A_precursor_in_A]
```

Fig. 6. Proposed publication / evidence ladder (not a results claim).

Paper	Focus	Dependencies
A (this draft)	Research Kit methods, Dual A precursor, hypnogram, Beacon n≈5	Ethics + 5 kits + pilot N
B	Phenotype vs intake / dentist labels; overnight band recalibration; optional Owens–Mayoral endotype / MAD-selective RMMA tests [17]	BruxScreen pathway [4]; ≥6 h nights; MAD setting log
C	Deeper EMG / Bruxoff / PSG-AV concordance	Larger N; lab PSG-AV

[PLACEHOLDER — OWNER: Koorosh] Preferred venue sequence (journal vs conference first).

Follow-on engineering: Harden iOS Dual A overnight (paired cues + unified export).

Directions for future investigation

See also the full audit checklist in

[PAPER_A_VALIDATION_AND_FUTURE_WORK.md](#) §7.

1. **True overnight corpus** — consented temple nights with wear ≥ 6 h (goal 8 h); regenerate Fig. 5; recalibrate band cutoffs.
 2. **Publishable algorithm appendix** — filter SOS/order, state-machine thresholds, SpO₂ curve coefficients; Python↔Swift epoch parity report.
 3. **Signal QC endpoints** — IR-DC coupling in-range fraction (10M–70M raw class), green SNR post-bandpass, SpO₂ quality-gate rate, BLE gap statistics (Table II).
 4. **Motion / orofacial confounding** — ACC-informed rejection of false tonic/phasic; compare to awake PPG+ACC chewing literature [3] without conflating goals.
 5. **Literature refresh** — re-verify Li ambulatory gap for temple optical devices published after Dec 2024.
 6. **Clinical labels (Paper B)** — BruxScreen-Q/C vs device-inferred load fractions; chairside hypnogram usability. Natural home for Owens & Mayoral (2026) endotypes and homeostatic-latency operationalization [17] — not Paper A endpoints.
 7. **Deeper EMG / PSG-AV (Paper C)** — expand Dual A precursor (in Paper A) to Bruxoff/PSG-AV RMMA subsets.
 8. **Ethics / data governance** — Beacon + optional McGill REB; DPA before raw log exchange; COI text finalized.
 9. **Hardware enablement** — charge-to-temple + **5 Research Kits to Pedro by 31 Aug 2026**; publication photos replacing placeholder SVGs (FIG-CO-012/016/026–029).
 10. **iOS Dual A overnight** — harden dual central + paired export after kit handoff.
-

Acknowledgments

[PLACEHOLDER — OWNER: All] Beacon Hospital, McGill / RI-MUHC, hardware partners (as appropriate), funding statements.
Do not include investor valuation language.

Conflicts of Interest

[PLACEHOLDER — OWNER: Koroosh] Draft disclosure: Research Associate, McGill / RI-MUHC; officer role at Dianyx Innovations (if still accurate) — **this manuscript concerns Orable / JAC extraoral temporalis work under McGill academic collaboration.** Any later complementary Dianyx commercial path is separate from Paper A inventorship.
[PLACEHOLDER — OWNER: John] Inventor / JAC Dental Solutions Limited; US provisional on Temporalis OMG path (patent pending — no claim text).
[PLACEHOLDER — OWNER: Ed/Pedro] Clinical practice disclosures / Beacon affiliations.

Ethics Statement

[PLACEHOLDER — OWNER: Ed/Pedro + Koorosh] IRB / REB protocol numbers, consent, and whether data are wellness pilot vs interventional study.

Data and Code Availability

[PLACEHOLDER — OWNER: All] Policy before sharing raw logs: de-identification, partner agreements, and whether algorithms are described at methods level only vs open-source release.
Internal references (not public): `src/analysis/overnight_states.py`, `OrableCore` processors, overnight report scripts [6], [7].

References

1. Li C, Yap S, Loh A, Yap YJ, Kujan O, Balasubramaniam R. Ambulatory devices to detect sleep bruxism: a narrative review. *Aust Dent J*. 2024;69(1 Suppl):S53–S62. doi:10.1111/adj.13057.
2. Nabavi S, Cogan J, Roy A, Canfield B, Kibler R, Emerick C. Sleep Monitoring with Intraorally Measured Photoplethysmography (PPG) Signals. 2022 *IEEE Sensors*. doi:10.1109/SENSOR52175.2022.9967075. (Confirm author line with Koorosh; contrast-only vs Oratable extraoral temporalis.)
3. Papapanagiotou V, Diou C, Zhou L, *et al*. A novel chewing detection system based on PPG, audio, and accelerometry. *IEEE J Biomed Health Inform*. 2017;21(3):607–618. doi:10.1109/JBHI.2016.2625271.
4. Lobbezoo F, Ahlberg J, Verhoeff MC, *et al*. The bruxism screener (BruxScreen): Development, pilot testing and face validity. *J Oral Rehabil*. 2024;51:59–66. doi:10.1111/joor.13442.
5. Oratable collaboration note — McGill / RI-MUHC academic engagement; extraoral temporalis anatomy; not Dianyx JV. Internal: [COLLAB_NABAVI_MCGILL.md](#). (Remove or replace with public URL before submission.)
6. Oratable algorithm architecture — Python ↔ Swift pipeline (Butterworth HR band, IR-DC LP, TFI/SASHB, overnight states). Internal: [../ALGORITHM_ARCHITECTURE.md](#). (Replace with citable methods appendix.)
7. Overnight night report — hypnogram-first bands and evaluable-night rules. Internal: [../OVERNIGHT_NIGHT_REPORT.md](#).
8. Product roadmap / Phase 0 temple vitals on Gen1. Internal: [../PRODUCT_ROADMAP.md](#).
9. Core ML training cohort plan (Tier 1 ≈ 20–30 users). Internal: [../CORE_ML_TRAINING_COHORT.md](#).
10. [PLACEHOLDER – OWNER: Koorosh] Charlton / pulse-wave analysis references as used in PPG morphology methods.
11. [PLACEHOLDER – OWNER: All] Additional SB / RMMA / ICAB–STAB normative cites for Introduction.
12. Cid-Verdejo *et al*. — Bruxoff vs PSG in OSA (see [BRUXOFF_PSG_GOLD_STANDARD.md](#) for full cite + claim discipline).
13. AcuPebble / Acurable HSAT landscape — nest with Arm P; Oratable ≠ AHI ([ACUPEBBLE_VS_ORALABLE_ANR.md](#)).
14. Happy Ring / Happy Health — Oura-like finger HSAT (K240236 monitoring; K242224 hAHI SaMD); nest class, not Oratable substitute ([HAPPY_RING.md](#)).

15. Orable Research Kit + Dual A Mac scripts — internal:
[ORABLE_RESEARCH_KIT.md](#) · [ANR_M40_CONCORDANCE.md](#) ·
[TEMPORALIS_ANATOMY_AND_PLACEMENT.md](#). (*Replace with citable methods before submission.*)
16. Kenhub. Muscles of mastication (tutorial / anatomy education). Temporalis anterior (elevate) vs posterior (retract) fiber actions. Used as anatomy teaching source for placement; publication figures FIG-CO-056 / FIG-CO-057 are Orable originals. (*Optional: add a journal-preferred textbook cite beside Kenhub.*)
17. Owens E, Mayoral P. Sleep bruxism: a novel theoretical perspective through the lens of the free energy principle (FEP). *Front Behav Neurosci*. 2026;20:1920406. doi:10.3389/fnbeh.2026.1920406. (*Clinical/neuro framing + Paper B; not Paper A methods. Bookmark: [ED_PEDRO_SB_FEP_DRAFT_PAPER.md](#).*)

Appendix A — Figure checklist (publication)

Fig	Asset / content	Status	Owner
1	Extraoral vs intraoral	Placeholder SVG (FIG-CO-011)	John + illustrator
2(a)–(b)	Temporalis function — anterior elevate / posterior retract	FIG-CO-056 / 057 Orable schematics (ready)	John — polish labels if venue requires
2(c)–(h)	Placement practice + device + Kit + ANR + Dual A wear + landscape	FIG-CO-003/012/016/026/031; landscape 027–029	John / Pedro
3	50 Hz pipeline	Mermaid + FIG-CO-007	Koorosh + John
4	IR-DC trough bout	Placeholder → real log	John
5	State hypnogram	FIG-CO-025 eng exemplar (~6 min)	John → pilot nights
6	Study ladder A→B→C	Mermaid in this draft	All
Table I–III	Cohort / QC / KPIs	Empty	Ed/Pedro + John

Appendix B — What this draft deliberately excludes

- Provisional patent claim text or claim charts
- Investor financials / valuation
- Dianyx product co-development framing

- Diagnostic claims (SB disorder, AHI, medical clearance)

End of Paper A working draft. Markup welcome as comments or tracked edits. Convert to IEEEtran after venue and authorship lock.